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Assessing the Capabilities of Large Language Models for Oil and Gas Industry Applications

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Abstract

Large Language Models (LLMs) are commonly evaluated on general-domain datasets such as MMLU or GSM8K. However, assessing their performance in specific domains requires creating a test set with domain-specific context. This work provides a comprehensive analysis of state-of-the-art LLM's capabilities with respect to oil and gas knowledge that has been conducted in the context of the EnergyAI project within ADNOC.

Our assessment is performed using two custom domain evaluation datasets. First, we measure the ability of the model to handle Multiple-Choice Questions (MCQ), which consists of questions that require a solid understanding and reasoning across upstream, midstream and downstream oil and gas operations. Second, we assess their ability to answer open-ended questions in the same domain, analyzing and quantifying the quality of the responses in relevance, accuracy and completeness. In addition to these evaluations, we examine other factors such as model size, throughput, response latency, and overall capabilities.

We have observed that the Llama3.1 405B model delivers a domain-specific performance comparable to proprietary models like GPT4o. Furthermore, models at a smaller scale - such as Llama 3.1 70B and Llama 3.2 90B - deliver excellent results relative to their size and offer an attractive performance-to-cost tradeoff. Nevertheless, all current LLM, including the most advanced open-source and proprietary models, exhibit notable limitations in their knowledge and reasoning capabilities within the oil and gas domain. These shortcomings highlight the need for targeted domain adaptation, deeper integration of technical knowledge, and rigorous evaluation tailored to the complexities of the industry.

Introduction

The rapid advancement of Large Language Models (LLMs) has created significant opportunities for their application across various industrial domains. While these models demonstrate remarkable capabilities on general knowledge tasks, their performance in specialized technical domains remains less understood. The oil and gas industry, with its complex technical terminology, specialized knowledge requirements, and critical decision-making processes, presents a particularly challenging domain for LLM evaluation.

Current LLM benchmarks such as Massive Multitask Language Understanding (MMLU) and Grade School Math 8K (GSM8K) provide valuable insights into general reasoning capabilities but fail to capture the nuanced requirements of domain-specific applications. In the petroleum industry, professionals must navigate complex technical concepts spanning upstream exploration and production, midstream transportation and processing, and downstream refining and marketing operations. Each of these segments requires deep technical knowledge, regulatory understanding, and the ability to integrate information from multiple sources.

The challenge of evaluating LLMs for oil and gas applications is compounded by the industry's reliance on both factual knowledge and complex reasoning. Professional certifications, such as those offered by the Society of Petroleum Engineers (SPE), require practitioners to demonstrate not only memorization of facts but also the ability to apply technical principles to solve real-world problems. This dual requirement, factual accuracy and applied reasoning, makes the oil and gas domain an ideal testbed for comprehensive LLM evaluation.

Recent developments in Retrieval Augmented Generation (RAG) - an AI framework for improving the quality of LLM-generated responses by grounding the model on external sources of knowledge to supplement the LLM's internal representation of information - offer promising approaches to enhance LLM performance in specialized domains. RAG implementation in an LLM-based question-answering system has two main benefits: it ensures that the model has access to the most current, reliable facts, and it provides users with the model's sources, allowing claims to be checked for accuracy and ultimately fostering trust. However, the effectiveness of these techniques in the oil and gas context has not been systematically evaluated. Furthermore, the trade-offs between model size, computational requirements, and domain-specific performance remain poorly understood, making it difficult for organizations to make informed decisions about LLM deployment.

This study addresses these gaps by providing a comprehensive evaluation framework specifically designed for the oil and gas industry. This work has been conducted in the context of EnergyAI project within ADNOC. Having a consistent and industry-specific evaluation framework is important to have an objective metric of the quality of the LLMs used across the Oil and Gas use cases. These evaluation datasets will be evolving to cover more custom and specific use-cases within the Oil and Gas domain.

Our approach combines quantitative assessment through multiple-choice questions with qualitative analysis of open-ended responses, providing a holistic view of LLM capabilities across different types of technical communication requirements.

36 different LLMs have been evaluated. The list of these models is provided below.

- **Microsoft (6)**: Phi-3-small-128k-instruct, Phi-3-mini-128k-instruct, Phi-3.5-mini-instruct, Phi-3-vision-128k-instruct, Phi-3.5-MoE-instruct, Phi-3-medium-128k-instruct.
- **Nvidia (4)**: Nemotron-Mini-4B-Instruct, Mistral-NeMo-Minitron-8B-Instruct, Nemotron-4-340B-Instruct, Llama-3.1-Nemotron-70B-Instruct-HF.
- **Mistral (5):** Mistral-7B-Instruct-v0.1, Mistral-7B-Instruct-v0.2, Mistral-7B-Instruct-v0.3, Codestral- 22B-v0.1, Mixtral-8x22B-Instruct-v0.1.
- **Meta (9)**: Llama-3.2-1B-Instruct, Llama-3.2-3B-Instruct, Llama-3-8B-Instruct, Llama-3.2-11B-Vision-Instruct, Llama-3.1-8B-Instruct, Llama-3.2-90B-Vision-Instruct, Llama-3.1-70B-Instruct, Llama-3.1-405B-Instruct-FP8, Llama-4 Scout.
- **Inception (4)**: jais-family-2.7b-chat, jais-adapted-7b-chat, jais-family-13b-chat, jais-adapted-70b-chat.
- **TII (2)**: falcon-11B, falcon-180B-chat.
- **Google (3)**: gemma-2-2b-it, gemma-2-9b-it, gemma-2-27b-it.
- **OpenAI (3)**: gpt-4o-mini_20241018, gpt-4-turbo_20241018, gpt-4o_20241018.

Evaluation Methodology

Our evaluation methodology employs a comprehensive multi-stage assessment approach designed to capture both the breadth and depth of LLM capabilities in the oil and gas domain. This framework addresses the multifaceted nature of technical communication in the industry, where professionals must demonstrate both precise factual knowledge and the ability to synthesize complex information.

Evaluation Datasets

Our primary evaluation utilizes two custom benchmark datasets specifically designed for the petroleum industry, MCQ and QA. We employ experts from the oil, gas, and energy sectors to obtain accurate and detailed responses for the benchmark datasets. All our benchmark datasets have reference answers (ground truth) written by domain experts.

MCQ Benchmark. A comprehensive dataset of 1210 multiple-choice questions covering the entire oil and gas value chain. Questions are categorized across upstream (exploration, drilling, reservoir engineering), midstream (transportation, processing, storage), and downstream (refining, petrochemicals, distribution) operations. The benchmark includes questions requiring both factual recall and complex technical reasoning, mirroring the rigor of professional certification examinations.

QA Benchmark. A curated set of 1300 open-ended questions designed to assess the model ability to provide comprehensive, accurate, and contextually appropriate responses to complex technical queries. These questions evaluate reasoning capabilities, technical communication skills, and the ability to synthesize information from multiple technical domains.

Evaluation Strategy

In our evaluation methodology we use an LLM to judge the quality of the generated answers with respect to the ground truth reference answers. This technique is known as LLM as a judge and provides a scalable evaluation process without relying on human annotators.

Generation of the responses. This approach uses the answers provided by the model to be evaluated by using a simple expert prompt that positions the model as a domain expert in oil, gas, and energy sectors. The strategy directly asks for detailed technical answers to user questions through straightforward single-step response generation. This method relies on the model's existing knowledge and training to provide comprehensive answers without additional reasoning scaffolding or multi-step processing.

LLM as a Judge Evaluation. LLM as a judge consists of using an LLM, like GPT-4, to evaluate the quality of the output produced by other models. Rather than relying on humans—which requires significant resources—this method automates and scales the evaluation process. This approach has gained traction as LLMs can assess nuances such as semantic similarity and contextual appropriateness that traditional metrics may not fully capture. Traditional metrics include:

- *Bilingual Evaluation Understudy* (BLEU), an algorithm that evaluates the quality of machine-translated text by measuring how closely it matches a professional human translation; the closer the machine translation is to a human's, the better the score. BLEU remains popular due to its high correlation with human judgments and cost-effectiveness.
- *Recall-Oriented Understudy for Gisting Evaluation* (ROUGE), which compares produced summaries or translations to reference (human-produced) versions. ROUGE is case-insensitive and is commonly used to assess the quality of automatic summarization and translation in natural language processing. By employing LLMs as evaluators alongside these metrics, the evaluation process benefits from both automated precision and human-like qualitative judgment.

We compute two metrics for the LLM as a Judge experiment. First, we ask GPT4 to provide a binary response if the answer is correct or not and we compute the percentage of correct responses per LLM. Secondly, we ask GPT4 to provide a score between 1 and 10, and request the LLM to assess their response based on the following criteria:

- Relevance: does it address the question appropriately?
- Accuracy: Is the information correct and consistent with the reference?
- Completeness: Does it cover the key points from the reference answer?
- Clarity: Is it well-expressed and easy to understand?

BLEU and ROGUE. BLEU and ROUGE are common metrics to evaluate Natural Language Generation tasks. BLEU measures the similarity between a generated text and reference text based on n-gram overlap, while ROUGE evaluates text quality by comparing overlaps of words or phrases, emphasizing recall and precision. For ROUGE we considered 1-ngram (ROUGE-1), 2-ngram (ROUGE-2) similarity and the longest common sequence (ROUGE-L).

Experiments and Results

In this section we are explaining the experiments conducted with the 36 evaluated LLMs and the obtained results with both the MCQ and QA eval datasets.

MCQ Evaluation Results

For each question, we are evaluating if the model provides an exact match to the correct answer, which will be 1 or 0; otherwise, using the prompt template listed in the Annex. We then compute the percentage of correct answers across all the MCQ questions.

The results are shown in Figure 1. The red dotted line indicates the expected percentage with random answers (baseline). The reason why some models are behind this baseline is because they are not following the instructions given in the prompt template correctly and are given a wrong or long answer to the multiple-choice question.

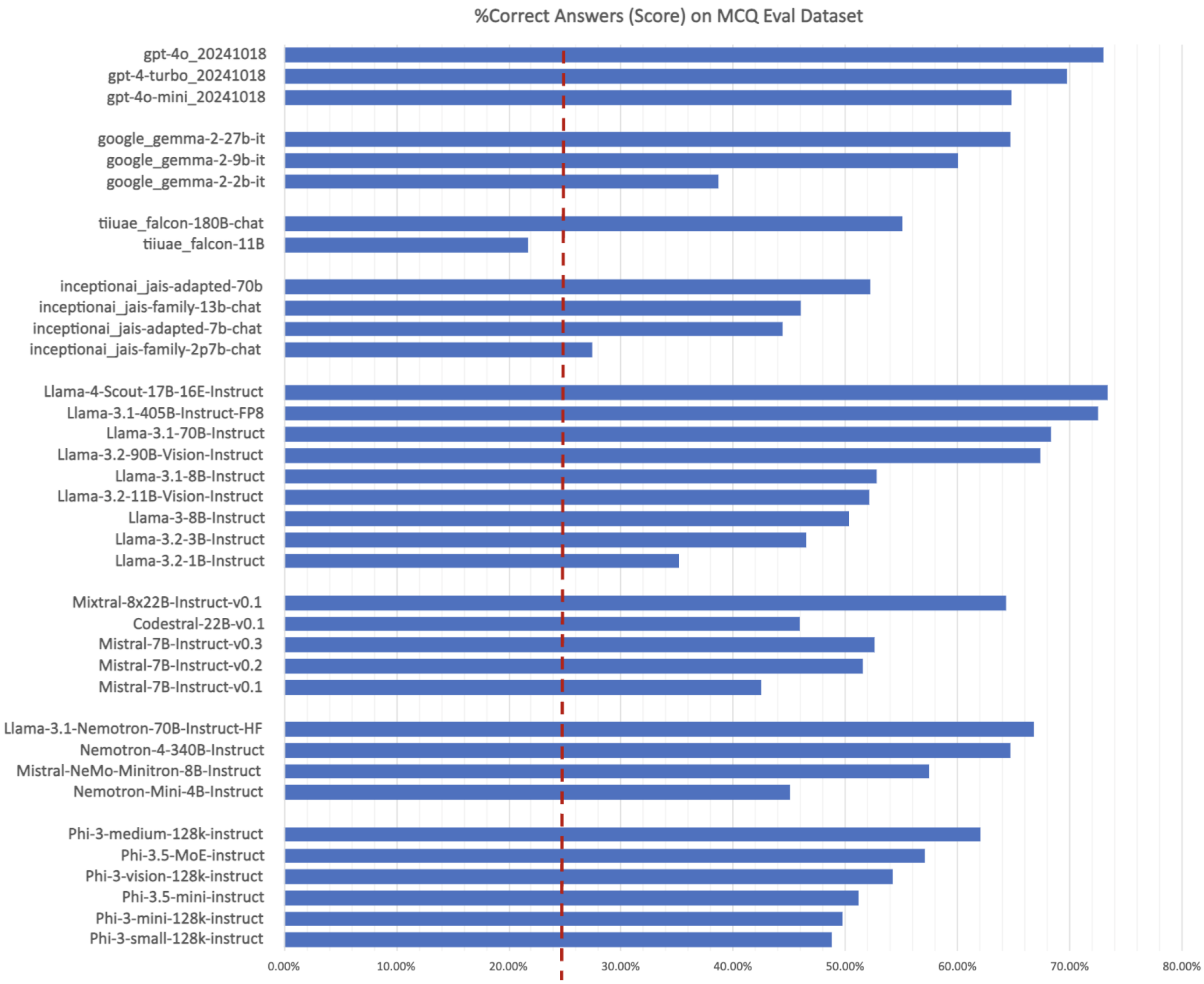

Figure 1—MCQ Evaluation Results

Evaluating Top Models and Mitigating Bias. To mitigate the impact of option order bias and reduce the potential for models to guess, we randomly shuffled the correct answers to create 10 distinct evaluation datasets. We then evaluated the top models on these 10 datasets, calculating the mean, median and standard deviation across the 10 experiments to ensure robust and unbiased results.

In Figure 2 we show the results of the top models and compare the mean results (10 shuffled rounds) and the initial ones (1 round).

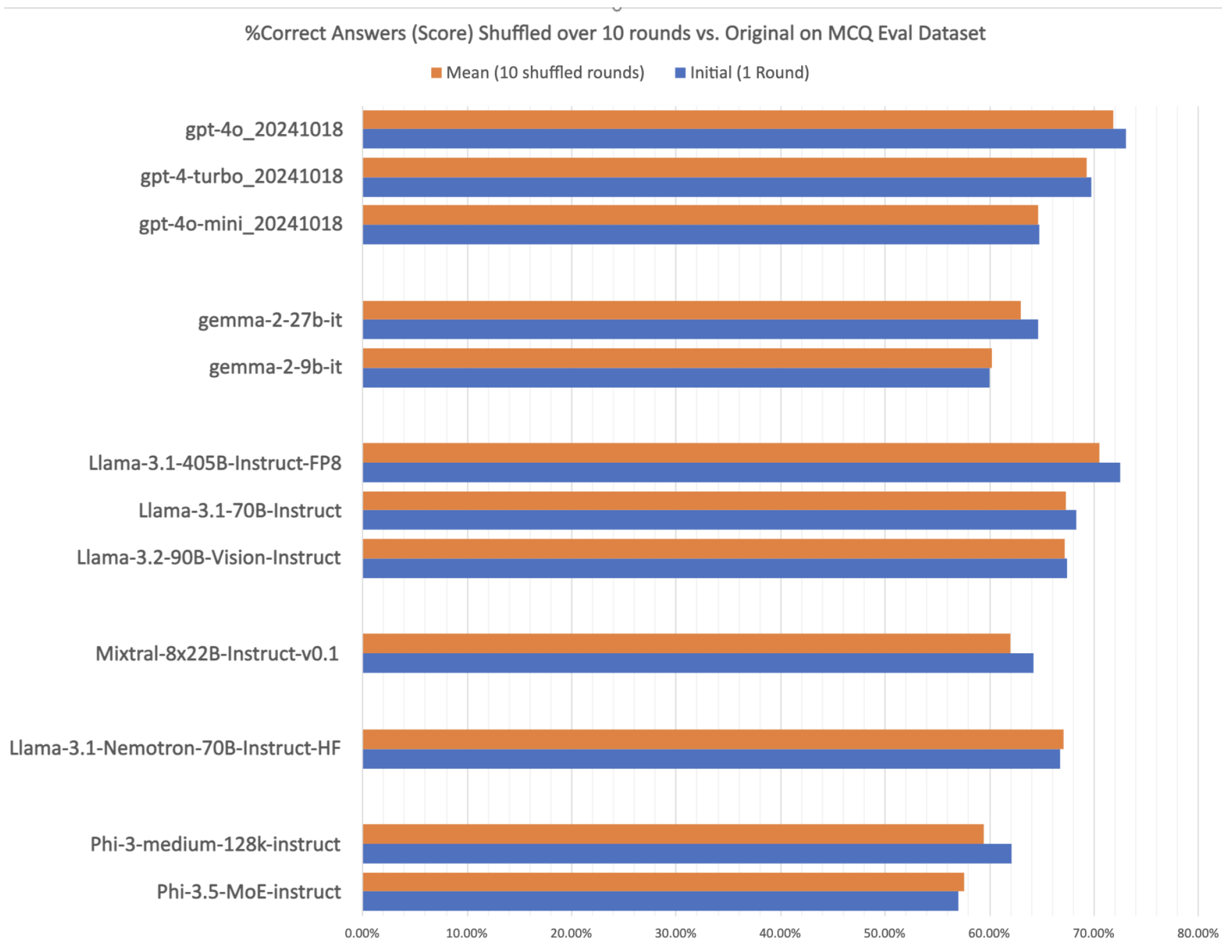

Figure 2—MCQ results over 10 shuffled rounds vs. original dataset

In general, we observe a slight percentage decrease in accuracy with the 10-rounds experiments versus the 1-round, mostly due to the order effect, as it has shown in table 1 under the column difference.

Table 1—MCQ eval results over 10 rounds vs. original dataset

Model	Initial (1 Round)	Mean (10 rounds)	Difference
Phi-3.5-MoE-instruct	57.10%	57.61%	-0.51%
Phi-3-medium-128k-instruct	62.10%	59.53%	2.57%
Llama-3.1-Nemotron-70B-Instruct-HF	66.83%	67.10%	-0.27%
Mixtral-8x22B-Instruct-v0.1	64.30%	62.03%	2.27%
Llama-3.2-90B-Vision-Instruct	67.43%	67.25%	0.18%
Llama-3.1-70B-Instruct	68.30%	67.34%	0.96%
Llama-3.1-405B-Instruct-FP8	72.55%	70.60%	1.95%
gemma-2-9b-it	60.02%	60.26%	-0.24%
gemma-2-27b-it	64.70%	63.03%	1.67%
gpt-4o-mini_20241018	64.83%	64.74%	0.09%
gpt-4-turbo_20241018	69.74%	69.34%	0.40%
gpt-4o_20241018	73.05%	71.94%	1.11%

As per the results, three models perform slightly better in the 10-round experiment than in the 1-round, the Phi-3.5-MoE-Instruct, the Nvidia-Llama-3.1-Nemotron-70B and the Gemma-2-9B, which suggests they had an effective post-training phase that follows the instructions accordingly.

The gpt-4-mini and the Llama-3.2-90B-Vision are the most stable ones, which suggest that these models are doing a much better work on following the instructions than the other models which are essentially more biased to either the 1-shot example in the prompt or the order of the correct answers.

MCQ Eval: Model Size vs. Accuracy. In figure 3 you can observe the evaluation results on the MCQ dataset per model size. We are excluding Llama3.1-405B and Nemotron-4-340B from this image for the sake of clarity. We refer the reader to the complete table listed in the appendix for more details. Model size has a direct impact on the amount of GPU that is required for the inference and on the latency cost. As we can observe the Llama3.1-70B and the Llama3.2-90B are options that are performing very well for their size.

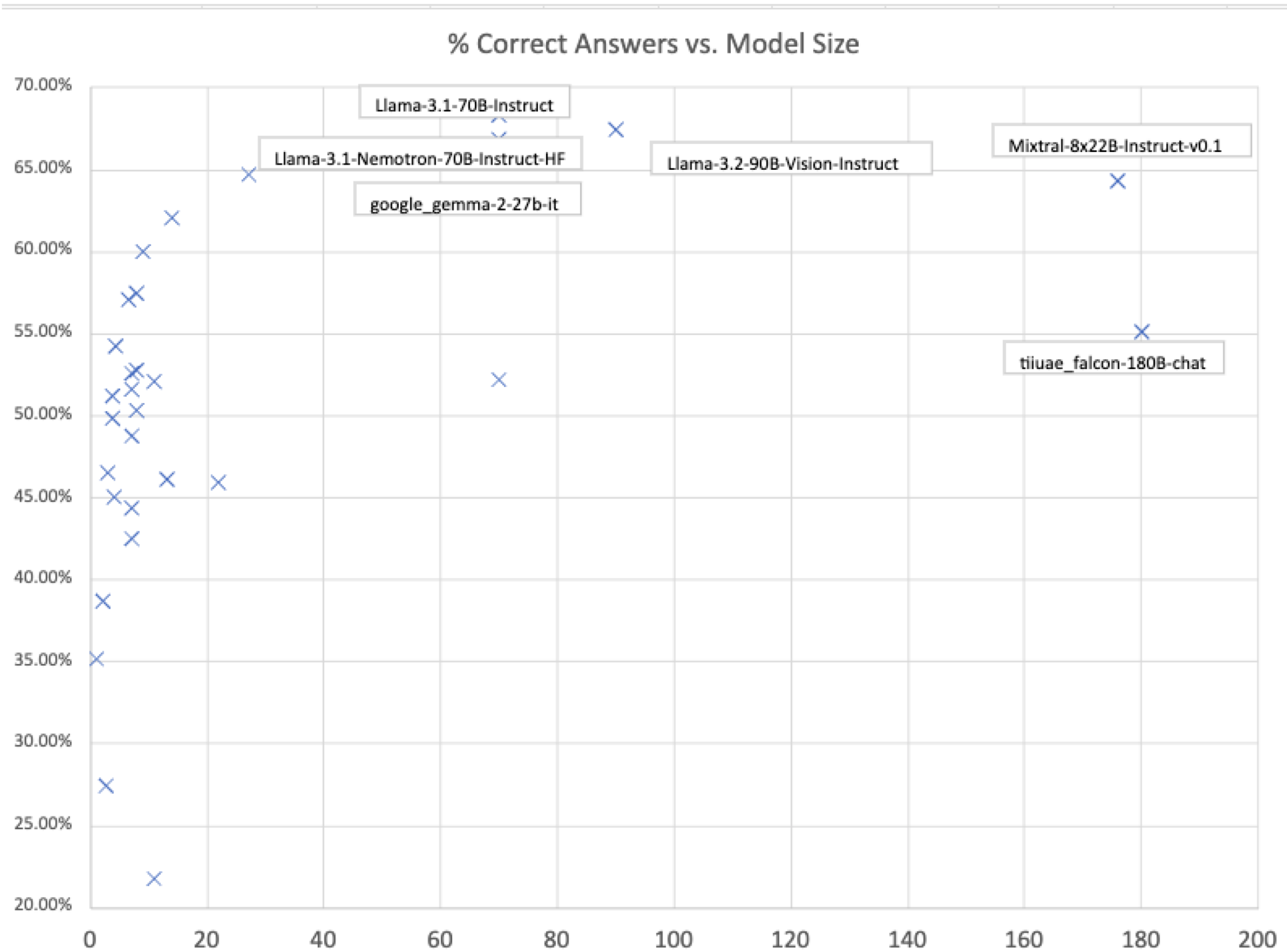

Figure 3—Correct answers per model size (number of parameters in Billion)

QA Results

In addition to evaluating the models using the MCQ evaluation dataset, we conducted experiments to assess their performance on natural language generation (NLG) tasks using 1300 question-answer pairs. For this evaluation, we considered multiple aspects and combined standard metrics for NLG tasks, such as Bilingual Evaluation Understudy (BLEU) [20] and Recall-Oriented Understudy for Gisting Evaluation (ROUGE) [21] scores with a human-like evaluation using GPT4 as the judge.

LLM as a Judge. The prompt templates that we have used are listed in the appendix of this report. We conducted the experiments for the top performing models based on the MCQ evaluations and the results are shown below.

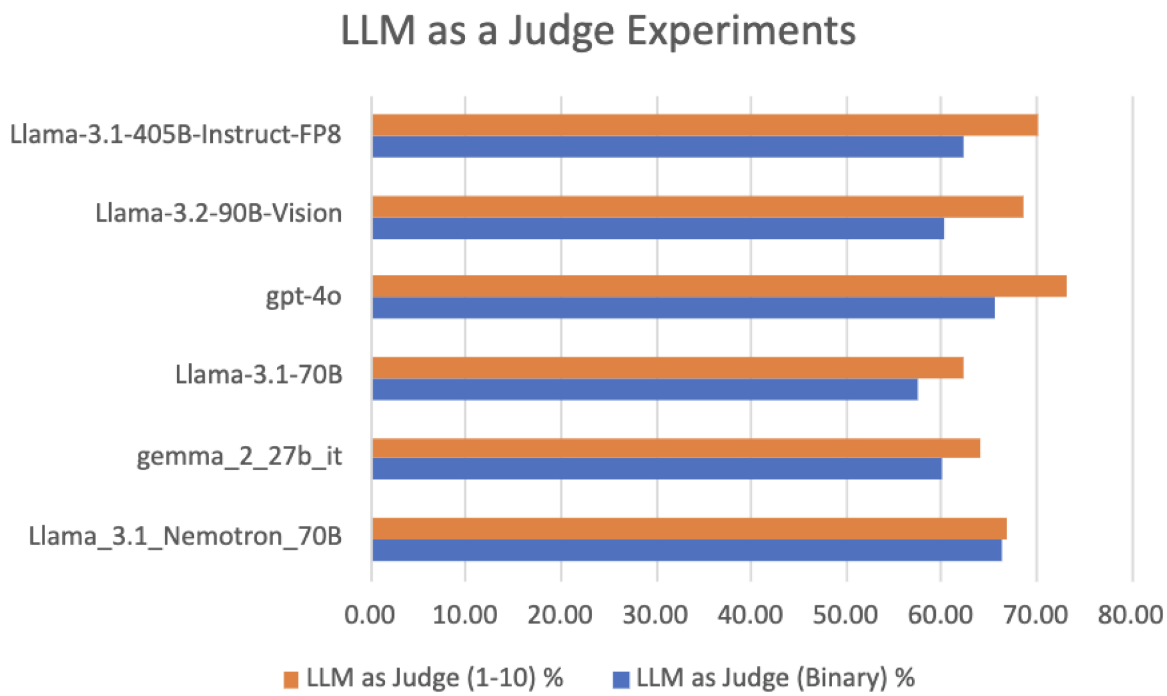

Figure 4—Chat Completion Results using LLM as a Judge Experiments

As we can see from the results Nvidia Llama3.1 Nemotron is the most stable one across the board, which highlights the improvements performed by the Nvidia team on the alignment conducted over the original Llama 3.1.

BLEU and ROUGE Metrics. As you can observe from the obtained results below, Llama3.2-90B-Vision has the highest percentage of similarity with the Ground-Truth answers that have been created and curated by humans. If we look at the similarity with GPT4o generated answers Llama3.1-405B has the highest score.

Table 2—BLEU and ROUGE experiments results. We have compared the responses to the questions of each model with the Ground-Truth (left part of the table) and with the answers to the questions generated by GPT4o (right part of the table)

	Ground truth Answers (Q-A) %				GPT4o Answers %			
Model	BLEU	ROUGE-1	ROUGE-2	ROUGE-L	BLEU	ROUGE-1	ROUGE-2	ROUGE-L
nvidia_Llama_3.1_Nemotron_70B	0.62	12.85	4	8.21	2.81	33.9	11.6	17.65
google_gemma_2_27b_it	0.65	16.38	4.55	10.36	3.21	39.66	12.74	20.86
Llama-3.1-70B	0.76	18.52	4.96	11.37	3.34	38.16	11.58	20.09
gpt-4o	4.61	29.44	11.8	20.45				
Llama-3.2-90B-Vision	**6.52**	**32.95**	**14.74**	**23.86**	12.16	52.33	25.02	34.78
Llama-3.1-405B-Instruct-FP8	6.3	32.62	14.32	23.68	**12.52**	**53.02**	**25.05**	**35.17**

Aggregated Score. We aggregated the scores from all the previous experiments into one metric to have a unified understanding of the chat completions quality. As the summary results in Figure 5 indicate, GPT4o achieved an aggregated score of **34.23%** (keeping into consideration we are not averaging with the GPT4o answers for the BLEU and ROUGE metrics) followed very closely by Llama3.1-405B with **33.53%** and Llama-3.2-90B-Vision with **33.13%**.

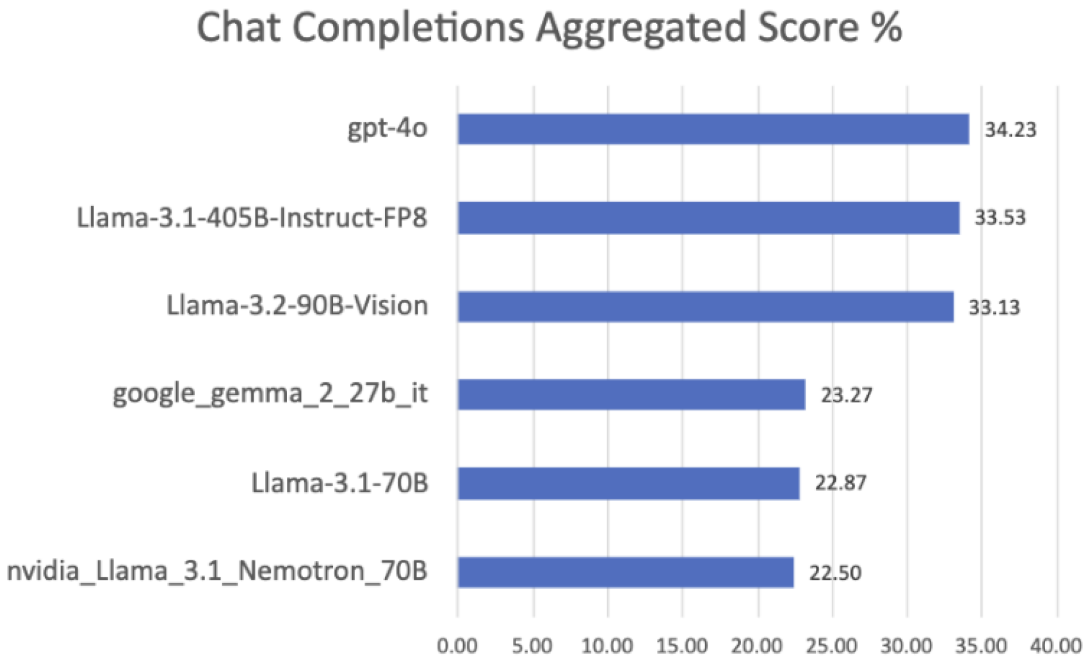

Figure 5—Chat Completion Aggregated Score across all the experiments

Conclusions

Our dual evaluation approach, combining domain-specific MCQs with qualitative analysis of open-ended responses, provides actionable insights on model capabilities across the entire oil and gas value chain. The study delivers practical performance metrics that enable organizations to make evidence-based decisions when selecting and implementing LLM solutions for technical oil and gas applications, balancing performance requirements against computational constraints and deployment costs.

This work has been conducted in the context of EnergyAI project within ADNOC to provide an objective evaluation of the LLM capabilities regarding Oil and Gas knowledge. It is extremely important to understand the capabilities and limitations of LLMs to cover specific domain use-cases and understand how it can be further improved.

As the landscape of frontier LLMs continues to evolve with enhanced capabilities and broader scope, this evaluation framework establishes a systematic foundation for ongoing assessment and benchmarking. The methodology developed through this research enables continuous monitoring of emerging models against oil and gas domain requirements, ensuring that the EnergyAI project can adapt to technological advances while maintaining rigorous performance standards.

Furthermore, this framework provides ADNOC with the analytical tools necessary to make informed decisions when selecting optimal LLMs for deployment in agentic AI systems, ensuring alignment between model capabilities and operational requirements across the organization's diverse technical applications.

Limitations

Despite the comprehensive nature of this evaluation, several limitations should be acknowledged that may affect the generalizability and interpretation of our findings.

1. Evaluation Methodology Constraints: The "LLM as a Judge" approach, while scalable, introduces potential circular bias by using GPT-4 to evaluate competing models. This methodology may not fully capture the nuanced domain expertise that human specialists would provide, and could inadvertently favor models with similar training paradigms to GPT-4. Additionally, traditional NLP (Natural

Language Processing) metrics like BLEU and ROUGE have known limitations in capturing semantic similarity and technical accuracy, particularly for specialized domain knowledge.

2. Dataset Scope and Representativeness: Our evaluation datasets, while substantial (1,210 MCQ and 1,300 QA pairs), represent a snapshot of oil and gas knowledge that may not fully encompass the breadth and evolving nature of the industry. The static nature of these datasets cannot account for rapidly changing industry practices, emerging technologies, or regional variations in terminology and procedures.
3. Model Coverage and Temporal Constraints: The evaluation of 36 models provides substantial coverage but represents a specific point in time in the rapidly evolving LLM landscape. Additionally, the evaluation focuses primarily on text-based interactions and does not thoroughly assess multimodal capabilities that could be crucial for real-world oil and gas applications involving diagrams, charts, and technical schematics.
4. Real-World Application Gaps: The evaluation primarily assesses models in isolated question-answering scenarios rather than the complex, multi-turn interactions typical of real-world professional applications. Critical aspects such as hallucination rates in safety-critical scenarios, model robustness to adversarial inputs, integration challenges with existing enterprise systems, and long-term reliability in production environments need to be explored.
5. Safety and Reliability Assessment: While accuracy is measured, the evaluation does not comprehensively assess model safety, particularly for applications where incorrect information could have significant operational or safety consequences.

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Appendix

MCQ prompt template

```
You are an expert in Oil, Gas, and Petroleum for certifications like Petroleum Engineering Certificate
(SPE).  You will be given Multiple Choice Questions, answer with the correct option number in angle
brackets "<" and ">".

Here is the question with the options:
########
{Question}
########
1. {Option1}
2. {Option2}
3. {Option3}
4. {Option4}

Instructions:
       - Just return the correct option number in "<" and ">"
       - Do not provide any other text

Example:    Question: What well radius should I use in model 2?
       1. 5 inches
       2. 10 inches
       3. 15 inches
       4. 17 inches
       Output: <1>
```

MCQ eval prompt template (binary)

```
Please act as an impartial judge and evaluate the quality of the response provided by an AI
assistant to the user question displayed below. Your evaluation should consider correctness and
helpfulness.

You will be given a reference answer and the assistant's answer.

Begin your evaluation by comparing the assistant's answer with the reference answer.
Identify and correct any mistakes. Be as objective as possible.

After providing your explanation, you must provide an integer score as 0 if the answer is incorrect
and 1 if the answer is correct by strictly following this format: <score>score </score>, for example:
<score> 0 </score>.
```

Question-Answer eval prompt template (1-10)

You are an expert evaluator in the gas, oil, and energy sector. Your task is to assess the quality of generated answers to technical questions by comparing them with reference answers. For each evaluation, you will be provided with:

- Question: The technical question.
- Reference Answer: The ideal or correct answer.
- Generated Answer: The answer to be evaluated.

Your instructions:
Evaluate and score the Generated Answer using both the Reference Answer and your own expert knowledge.

Assess the Generated Answer based on the following criteria:

- Relevance: Does it address the question appropriately?
- Accuracy: Is the information correct and consistent with the reference?
- Completeness: Does it cover the key points from the reference answer?
- Clarity: Is it well-expressed and easy to understand?

Score the Generated Answer on a scale from 1 to 10, where:

- 10: Excellent match in all aspects. Strictly follow this format when providing the score: <score>[score]</score>, for example: <score>0</score>.
- 1: Poor match; fails to meet the criteria.
- 0: Illegible or nonsensical text. Provide a very brief explanation for the score, highlighting the strengths and weaknesses.

Output Format:

- Score: <score>[1-10 or 0]</score>
- Explanation: [Your brief justification]

Example Evaluation:

Question:
What are the primary differences between upstream and downstream operations in the oil industry?

Reference Answer:
Upstream operations involve the exploration and production of oil and gas, including searching for underground or underwater oil fields and drilling wells. Downstream operations refer to the processes involved in refining crude oil, distributing, and selling the refined products like gasoline, diesel, and other petrochemicals to consumers.

Generated Answer:
Upstream is about finding oil underground, while downstream is turning oil into products we use every day.

Evaluation:

- Score: <score>7</score>
- Explanation: The Generated Answer correctly identifies the basic concepts of upstream and downstream operations but lacks detail. It mentions exploration (“finding oil underground”) and processing (“turning oil into products”), aligning with the Reference Answer, but does not elaborate on production, refining specifics, or distribution.

MCQ: a couple of examples

{"Question": A well has ceased flowing due to a buildup of saltwater in the tubing. The well is standing full of produced fluid with a density of 8.9 lbm/gal (1.067 SG). A pumping unit is to be installed to lift the fluid from a depth of 10,000 ft (3,048 m). Expected produced water rate is 220 bbl/D (35 m3/day). The pumping unit will be counterbalanced such that essentially all the sucker rod and fluid load will be offset by the counterweights. The pumping unit manufacturer requires a prime mover having 125% excess power to run the unit. What is the rated horsepower, hp (KW), of the prime mover needed to run the pumping unit?”,
"Options": {"1": "15 (11)", "2": "20 (15)", "3": "35 (26)", "4": "40 (30)"}, "Answer": "4"}

{"Question": "A well is to be fracture stimulated down the production casing with the following conditions: Fracture propagation gradient: 0.95 psi/ft (2.19 SG) Near well pressure losses: 1,000 psi (6,895 KPa) Casing friction pressure losses: 1,600 psi (11,032 KPa) Surface piping friction losses: 300 psi (2,068 KPa) Stimulation rate: 65 bbl/minute (10.33 m3/min) Perforations: 18,100 ft (5,517 m) MD/16,100 ft (4,907 m) TVD Fluid Density: 8.5 lbm/gal (1.02 SG) 50% Standby Horsepower on location Pump truck rated horsepower: 2,000/truck. Assume 85% Pump Efficiency How many pump trucks will be required?",
"Options": {"1": "11", "2": "14", "3": "16", "4": "17"}, "Answer": “3”}

Q-A: a couple of examples

Q: Why are beveled couplings or integral joint connections used in certain well designs?

A: These connections reduce sliding friction, which can be beneficial in wells with considerable borehole friction or unconsolidated formations.

Q: What are the formation mechanisms that allow for the existence of gas traps in the ultra-shallow layers of a basin, and how do they differ from those in deeper gas reservoirs?

A: The formation mechanisms for gas traps in the ultra-shallow layers of a Basin are influenced by sedimentary processes, structural features, and biogenic gas generation, differing significantly from the thermogenic processes and complex structures seen in deeper reservoirs.

Model Size vs. MCQ evaluation

In the table below you can see the % of MCQ correct answer per model size

Model	Params (B)	%Correct Answers
Phi-3-small-128k-instruct	7	48.80%
Phi-3-mini-128k-instruct	3.8	49.80%
Phi-3.5-mini-instruct	3.8	51.20%
Phi-3-vision-128k-instruct	4.2	54.20%
Phi-3.5-MoE-instruct	6.6	57.10%
Phi-3-medium-128k-instruct	14	62.10%
Nemotron-Mini-4B-Instruct	4	45.09%
Mistral-NeMo-Minitron-8B-Instruct	8	57.50%
Nemotron-4-340B-Instruct	340	64.7%
Llama-3.1-Nemotron-70B-Instruct-HF	70	66.83%
Mistral-7B-Instruct-v0.1	7	42.48%
Mistral-7B-Instruct-v0.2	7	51.60%
Mistral-7B-Instruct-v0.3	7	52.61%
Codestral-22B-v0.1	22	45.95%
Mixtral-8x22B-Instruct-v0.1	176	64.30%
Llama-3.2-1B-Instruct	1	35.17%
Llama-3.2-3B-Instruct	3	46.49%
Meta-Llama-3-8B-Instruct	8	50.30%
Llama-3.2-11B-Vision-Instruct	11	52.10%
Meta-Llama-3.1-8B-Instruct	8	52.81%
Llama-3.2-90B-Vision-Instruct	90	67.43%
Llama-3.1-70B-Instruct	70	68.30%
Llama-3.1-405B-Instruct-FP8	405	72.55%
Llama-4-Scout-17B-16E-Instruct	109	80.93%
inceptionai_jais-family-2p7b-chat	2.7	27.45%
inceptionai_jais-adapted-7b-chat	7	44.39%
inceptionai_jais-family-13b-chat	13	46.09%
inceptionai_jais-adapted-70b	70	52.20%
tiiuae_falcon-11B	11	21.74%
tiiuae_falcon-180B-chat	180	55.10%
google_gemma-2-2b-it	2	38.68%
google_gemma-2-9b-it	9	60.02%
google_gemma-2-27b-it	27	64.70%